

# Engineering Curves: Construction of Cycloid

Rolling Circle Method with Tangent & Normal

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Gajavada Sanjeevkumar

# Problem Statement

## Question

Draw a **cycloid** generated by a circle of diameter  $D = 50$  mm rolling along a straight base line for **one complete revolution** without slipping.

Construct a **tangent** and a **normal** to the cycloid at any point  $P$  located on the curve.

## Step 1: Draw Generating Circle & Directing Line

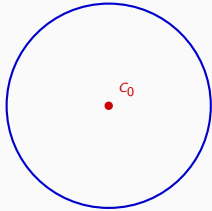
$C_0$  → Initial Center Point

$AA'$  → Directing Line

**Axis** → Center Line



## Step 1: Draw Generating Circle & Directing Line

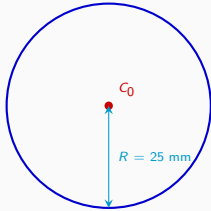


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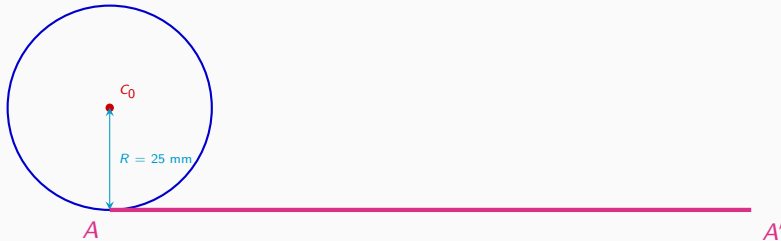
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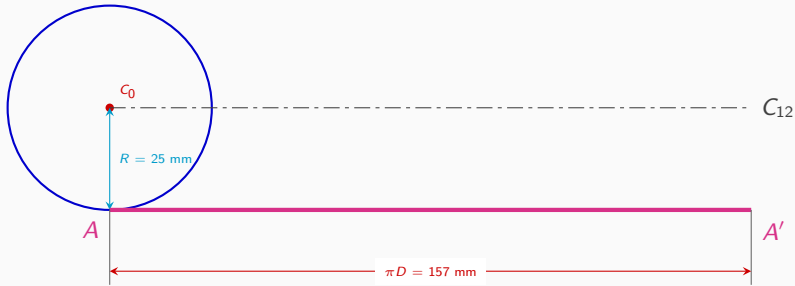
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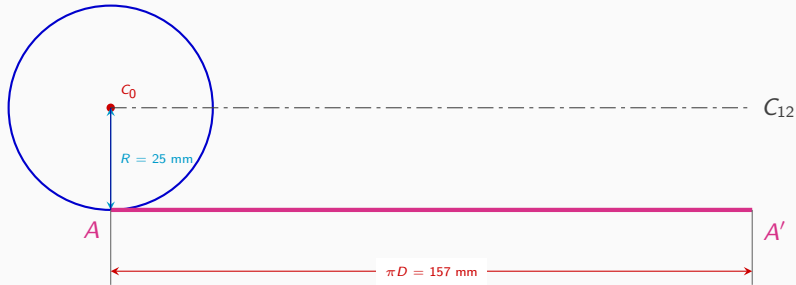
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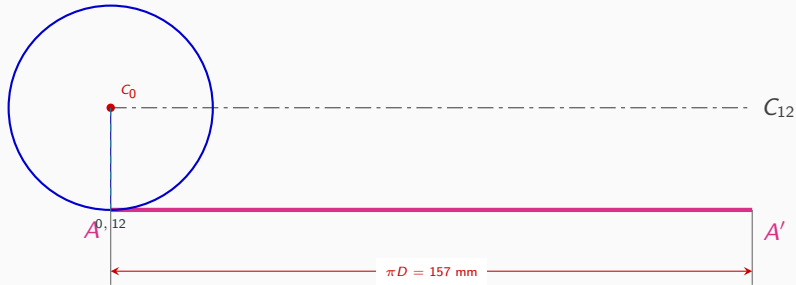
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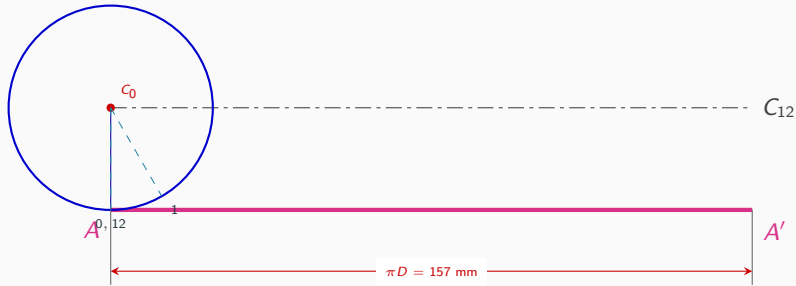
## Step 2: Divide Generating Circle into 12 Equal Parts

$AA'$  → Directing Line  
**Axis Line** → Center Line  
 $0 \dots 12$  → Circle Divisions



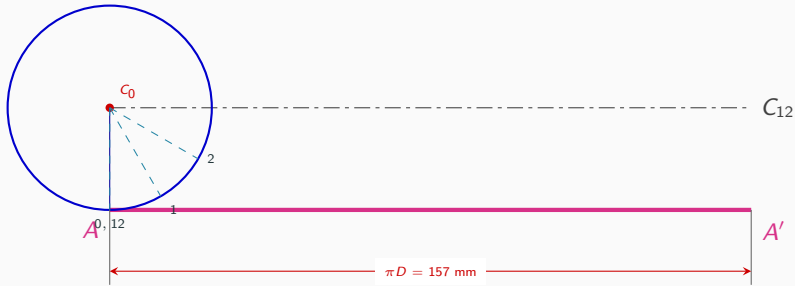
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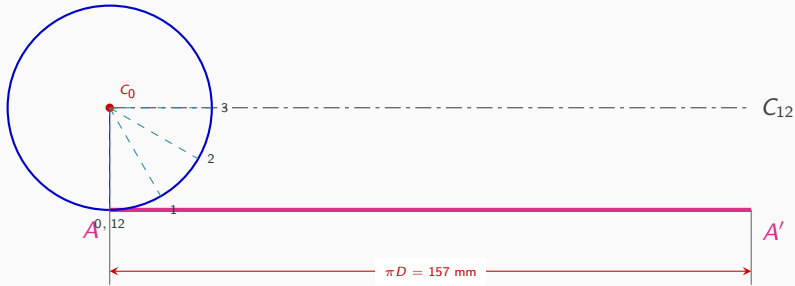
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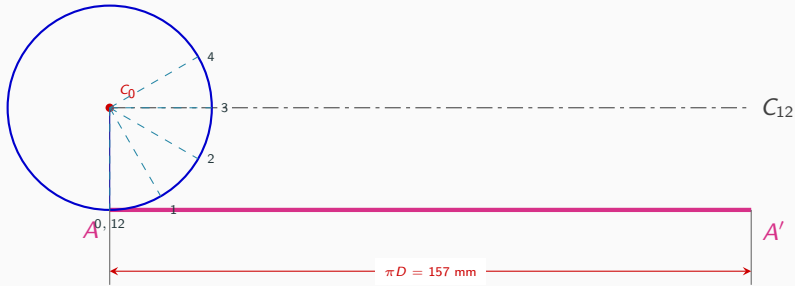
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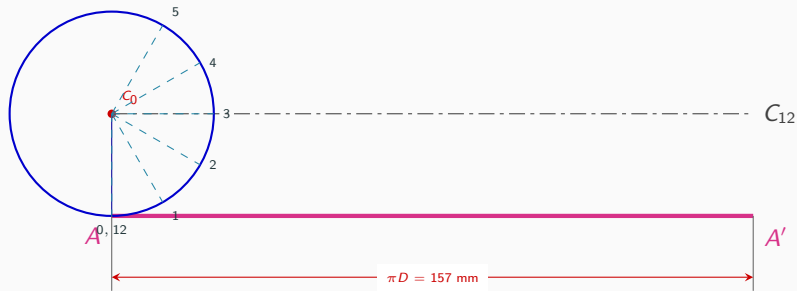
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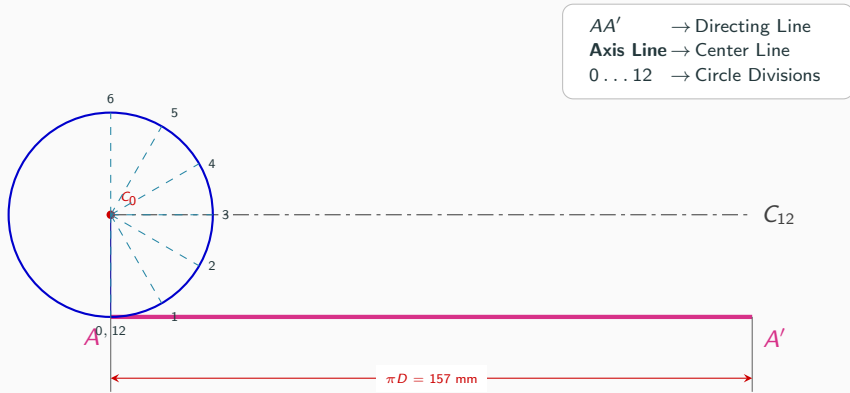




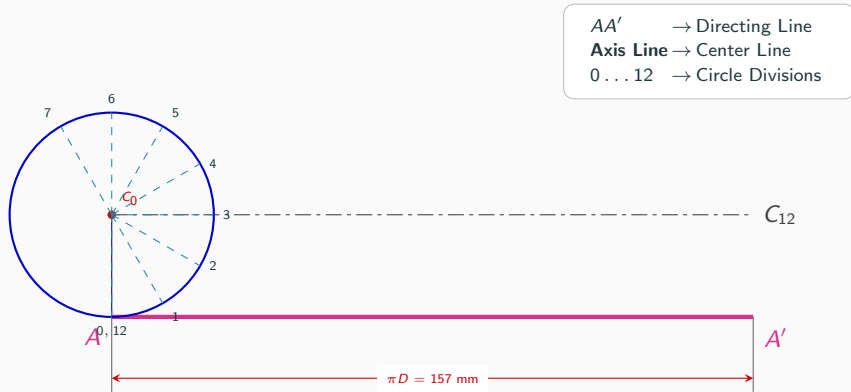
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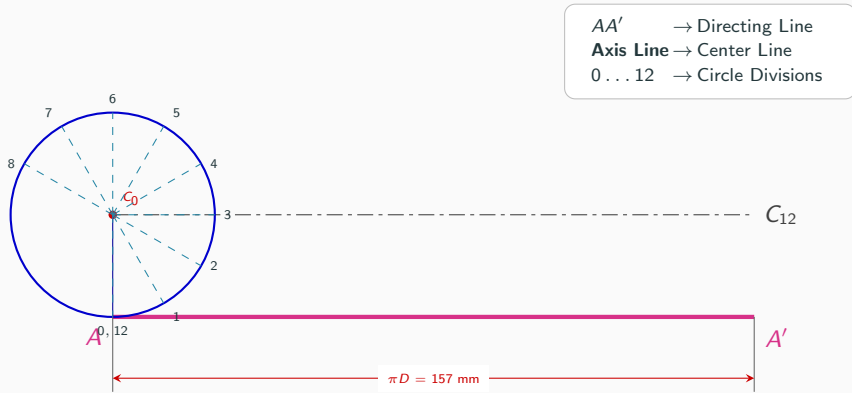
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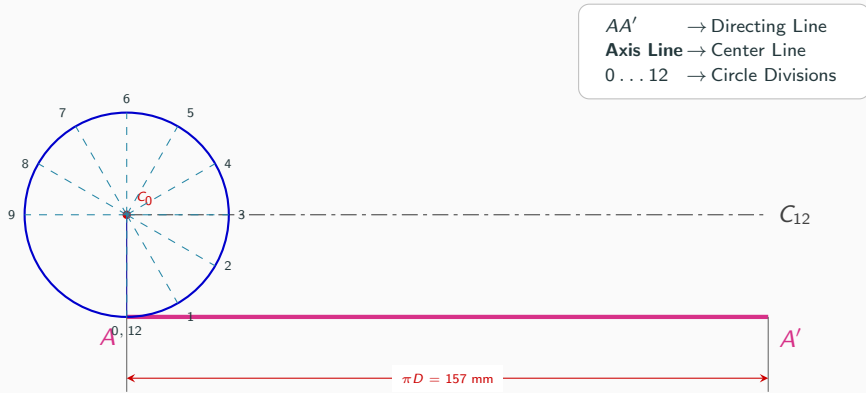
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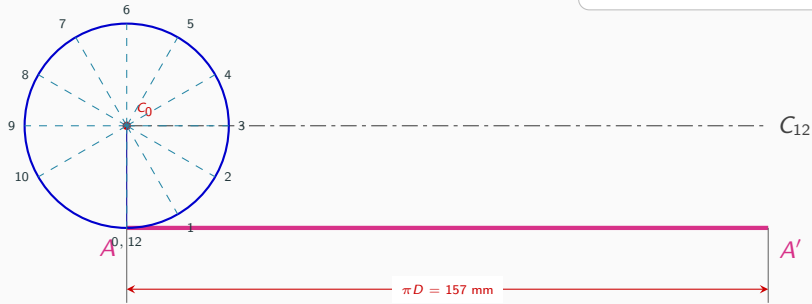
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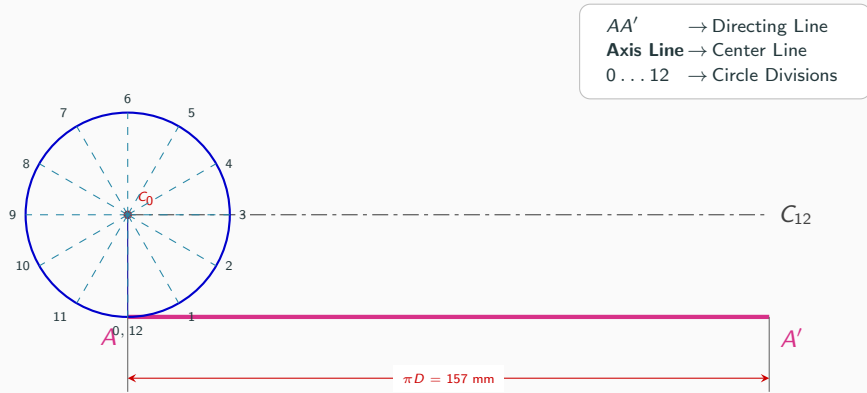
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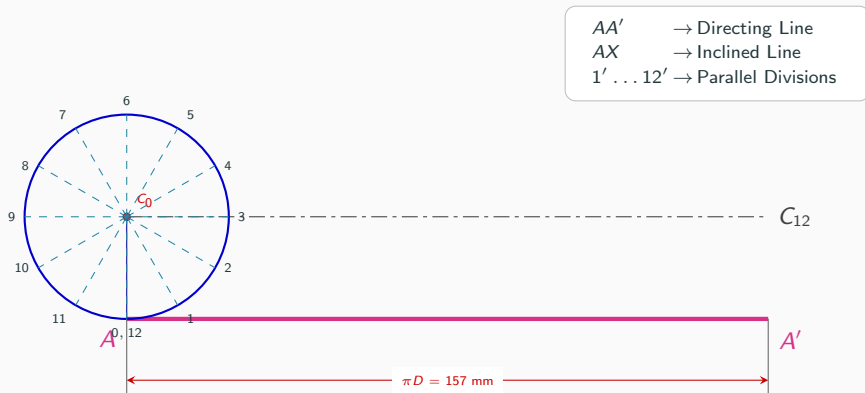
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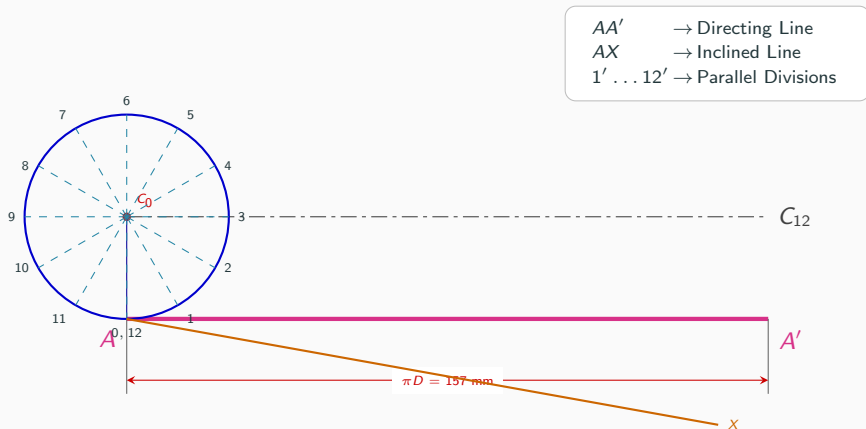


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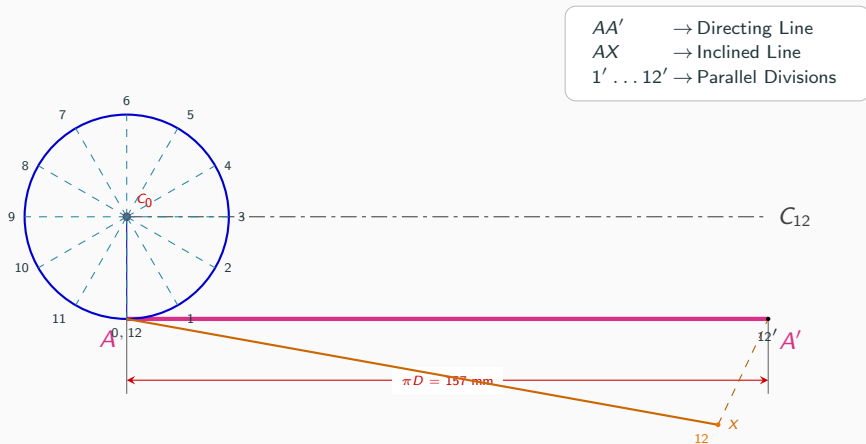




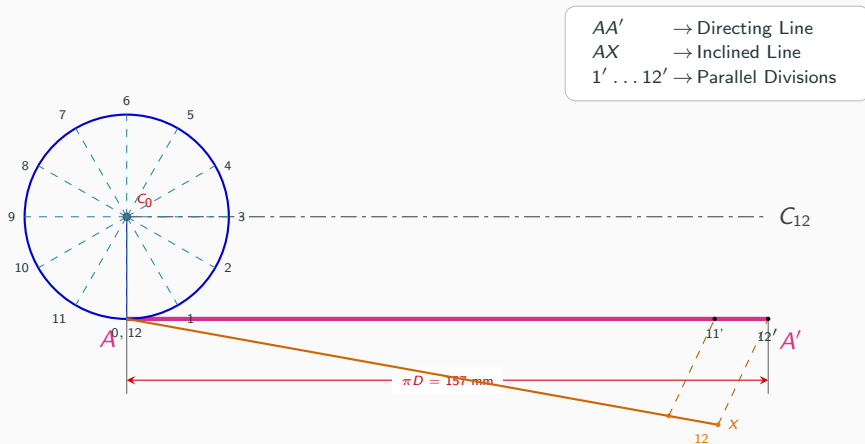
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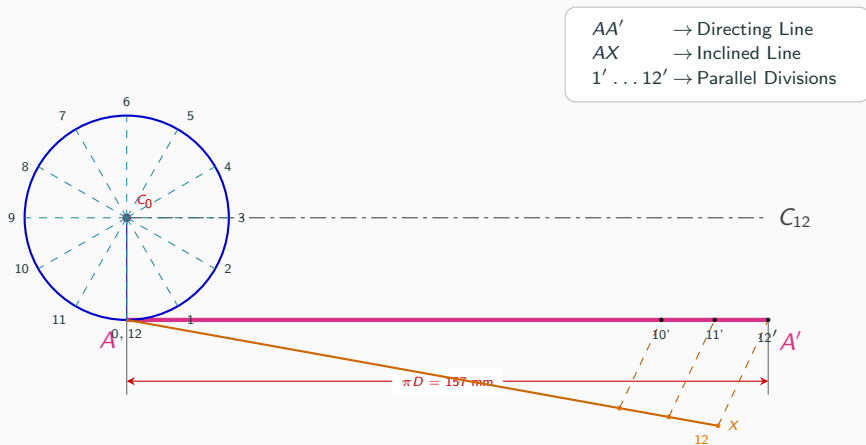
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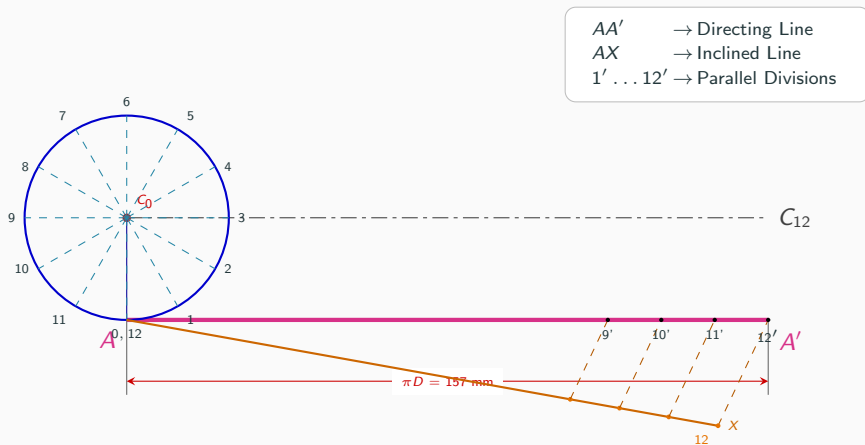
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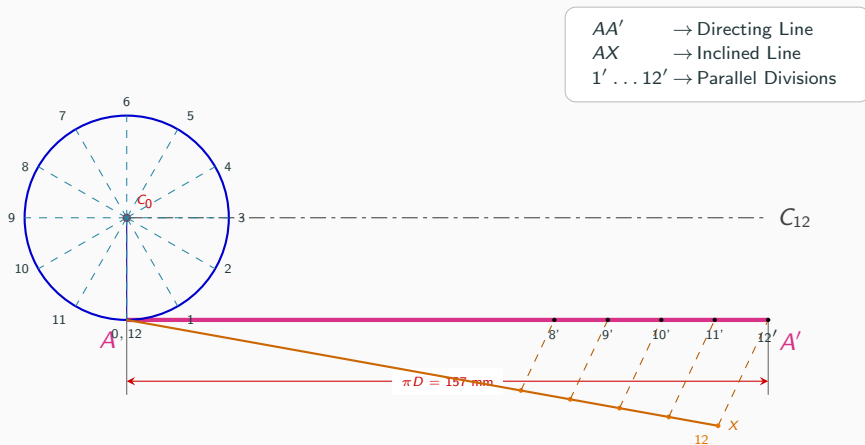
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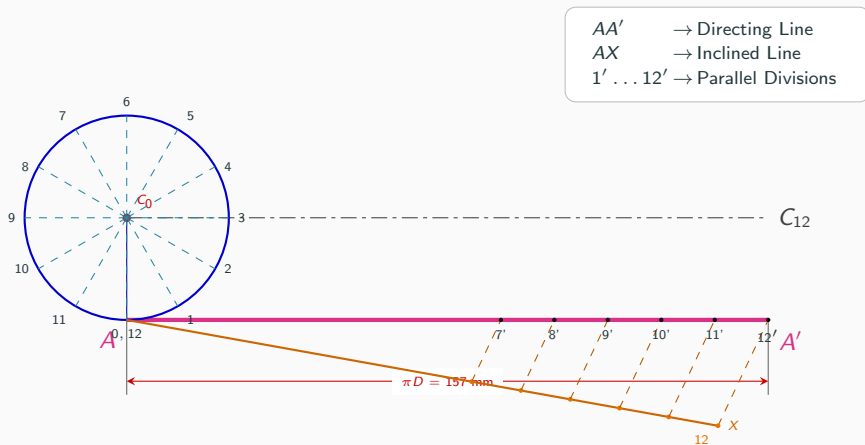
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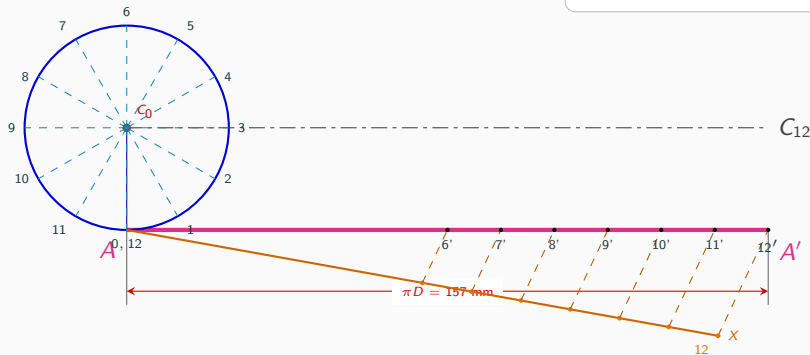


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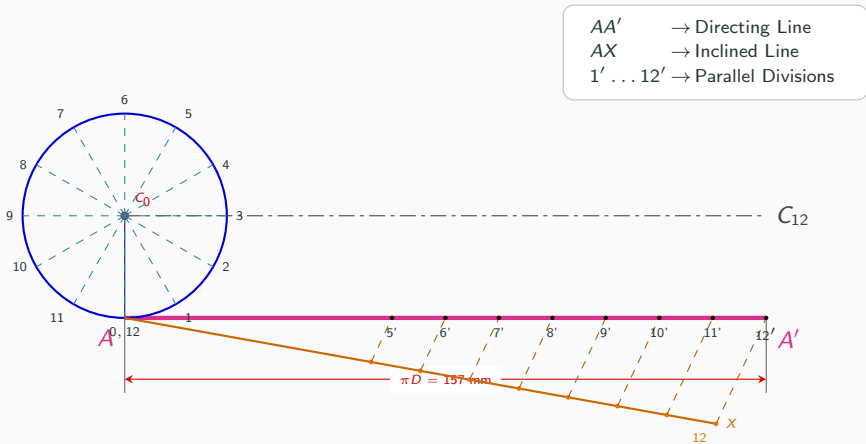
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$AA'$  → Directing Line  
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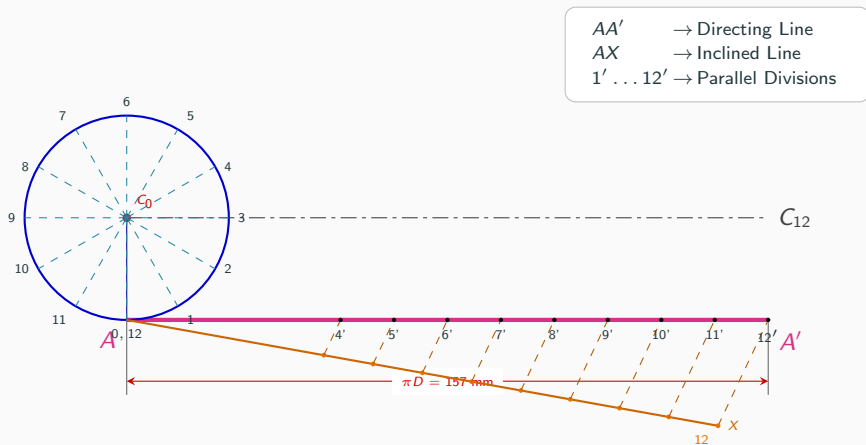




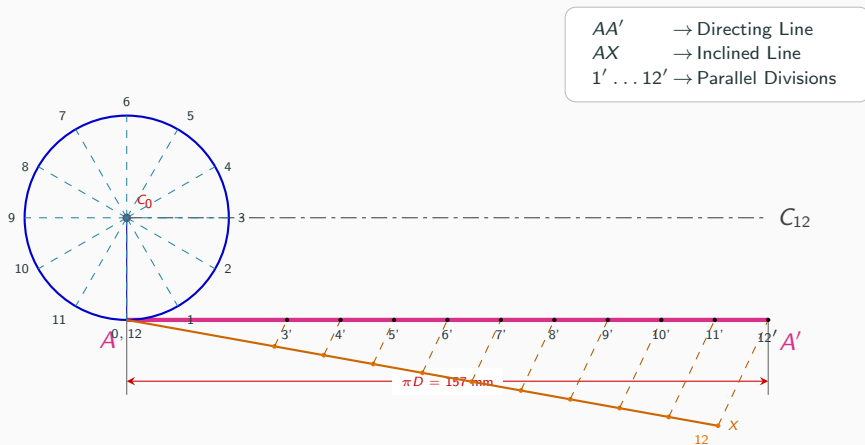
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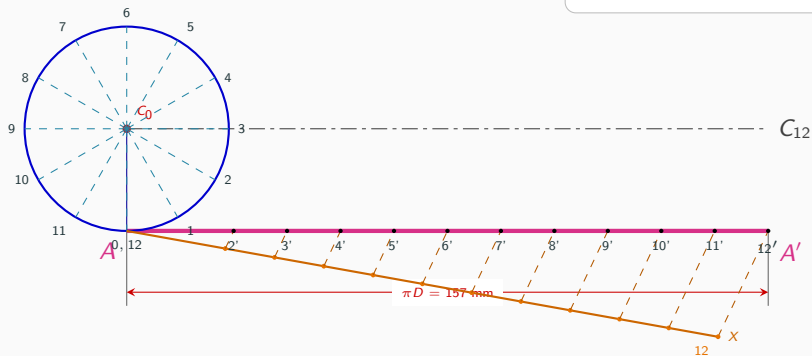


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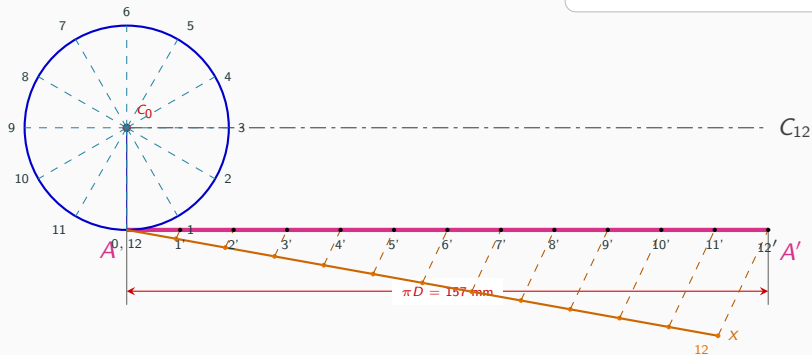
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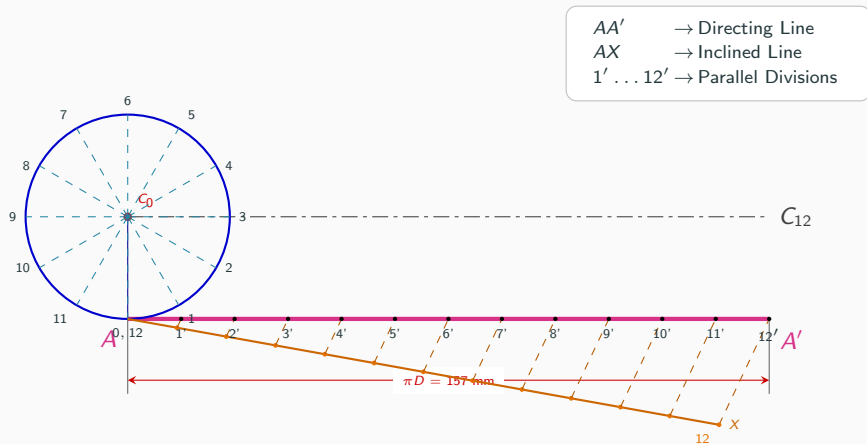


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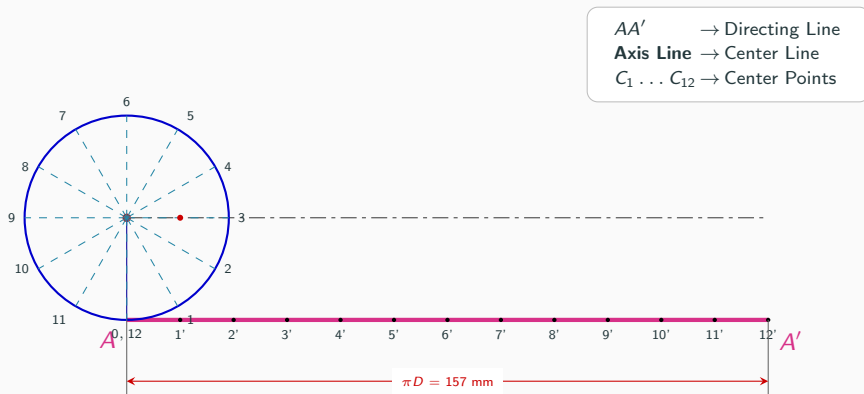
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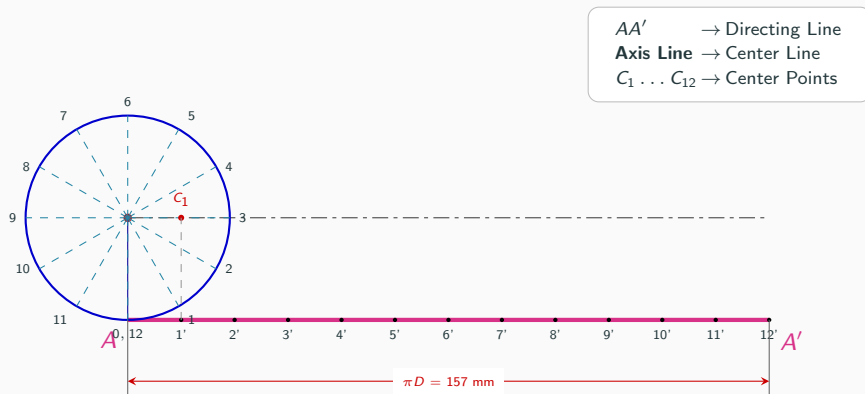
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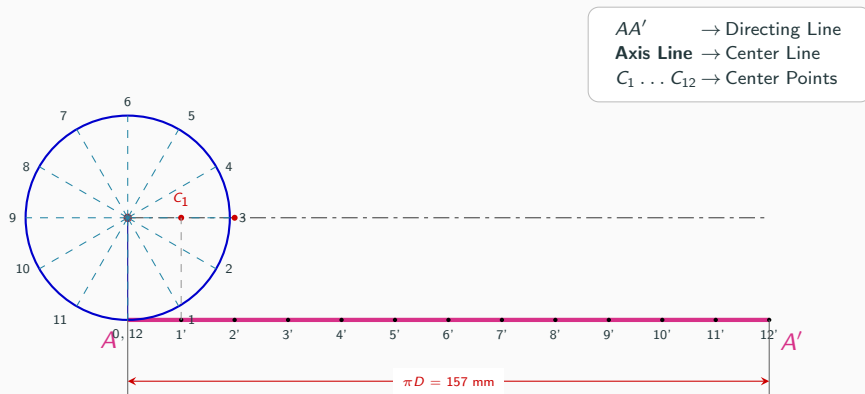


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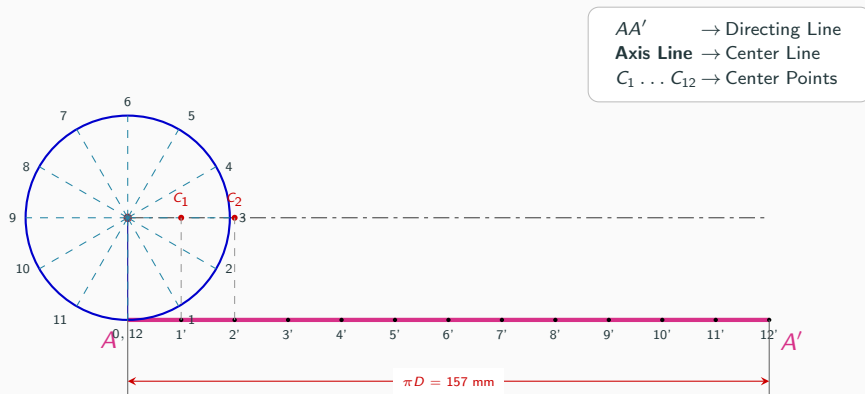




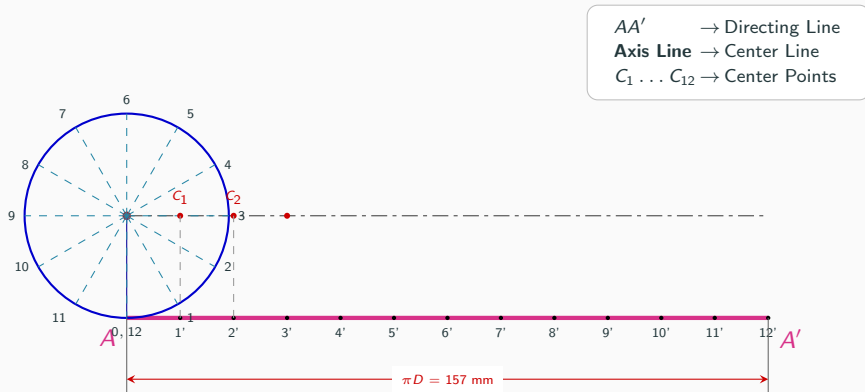
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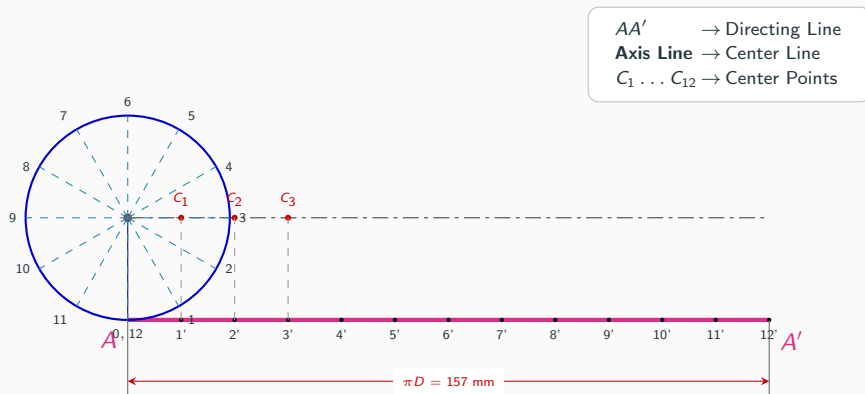
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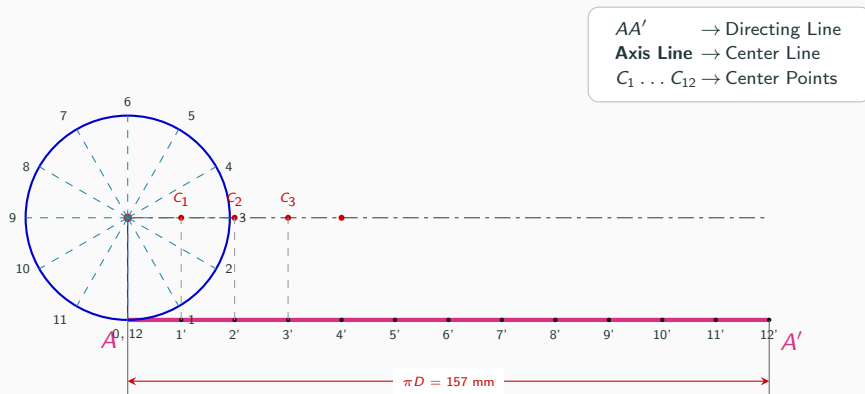
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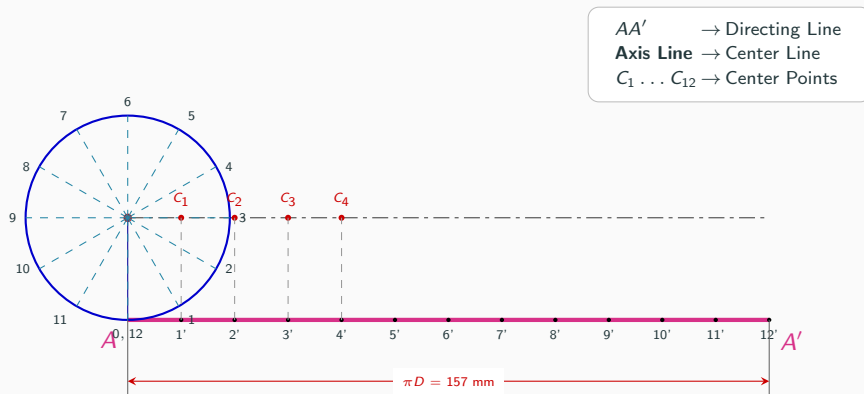
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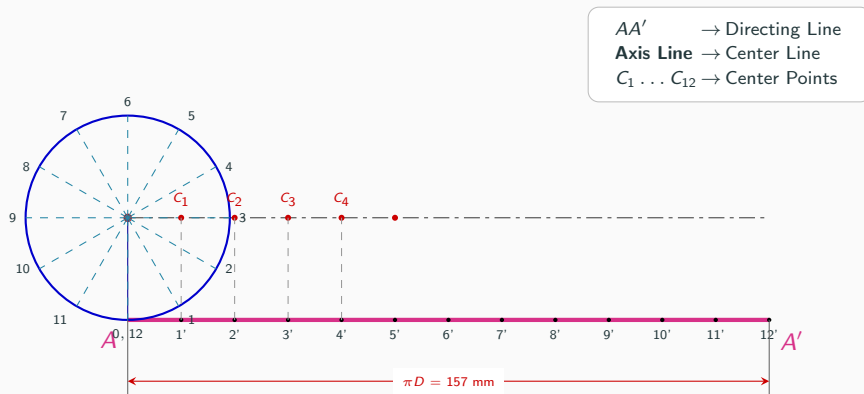
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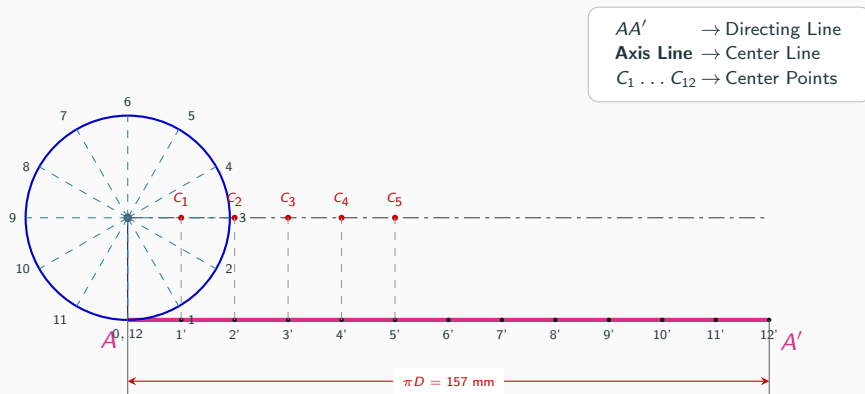
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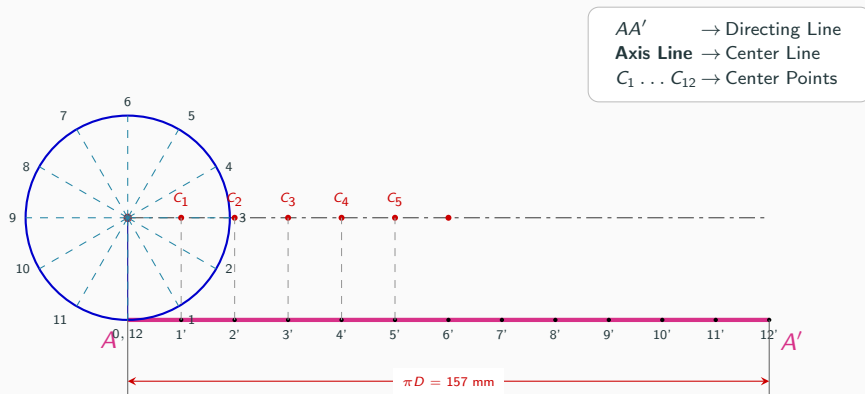


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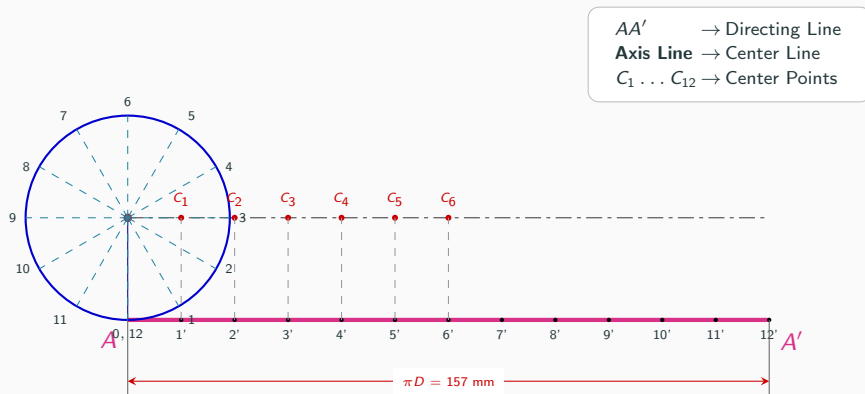




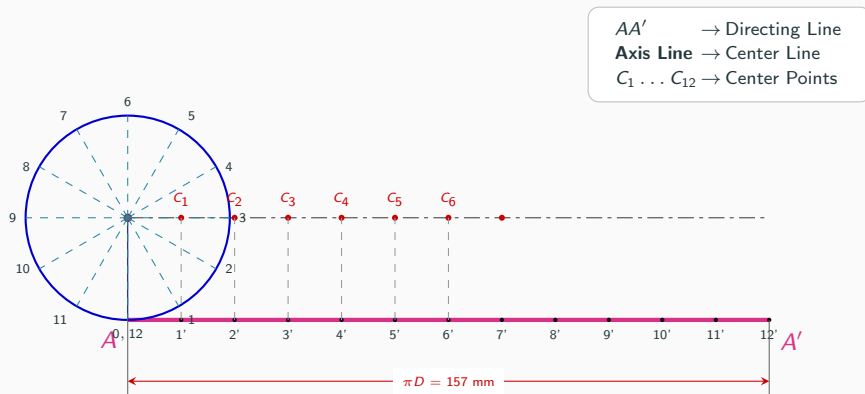
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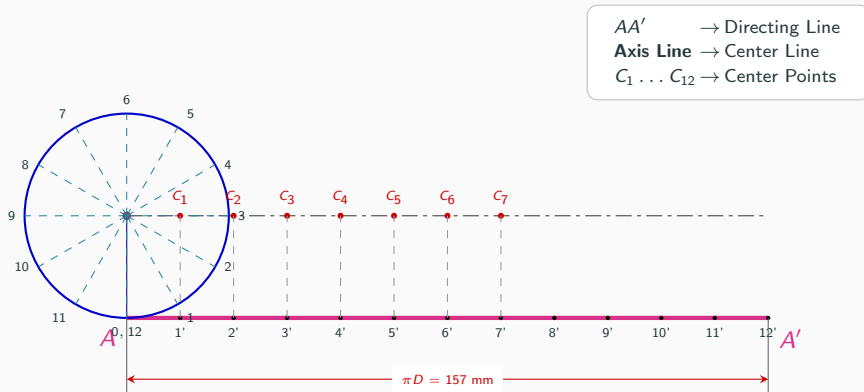
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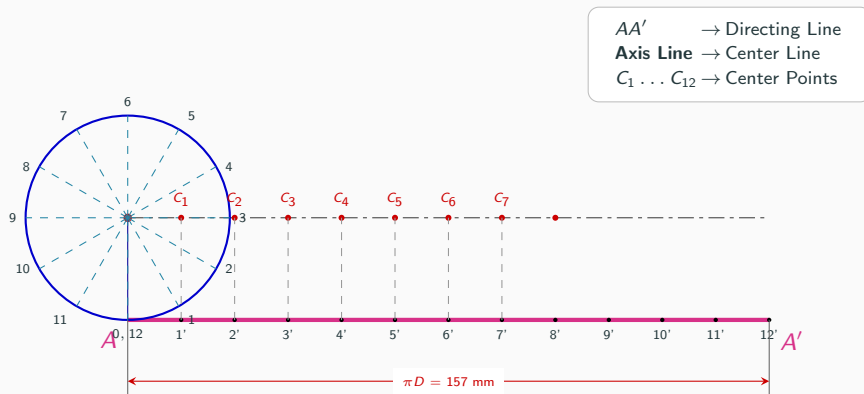
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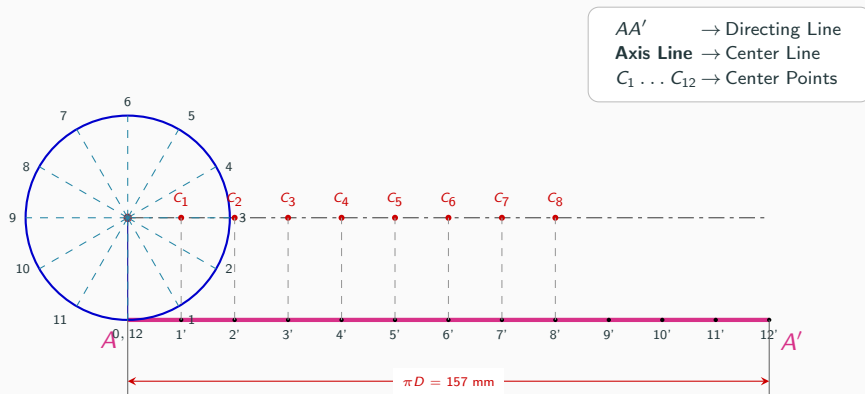
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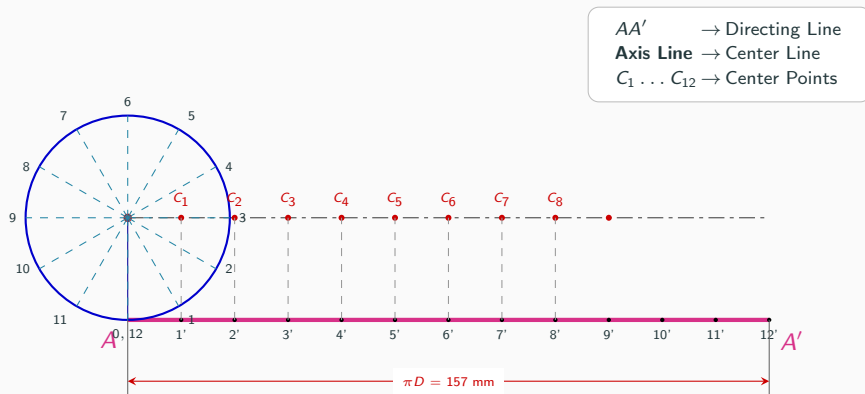
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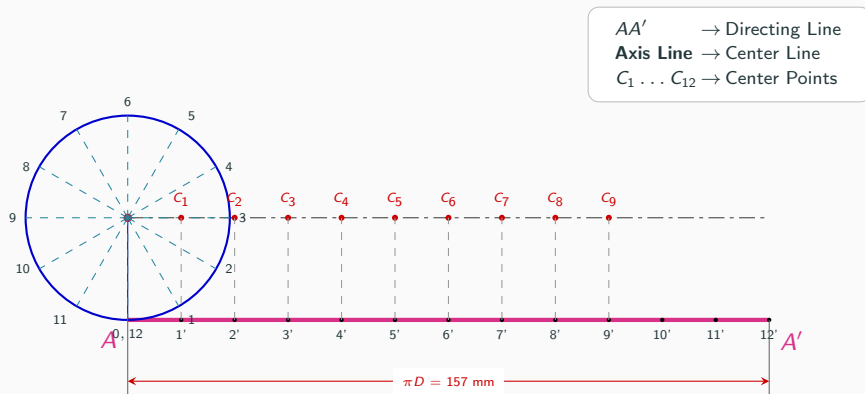
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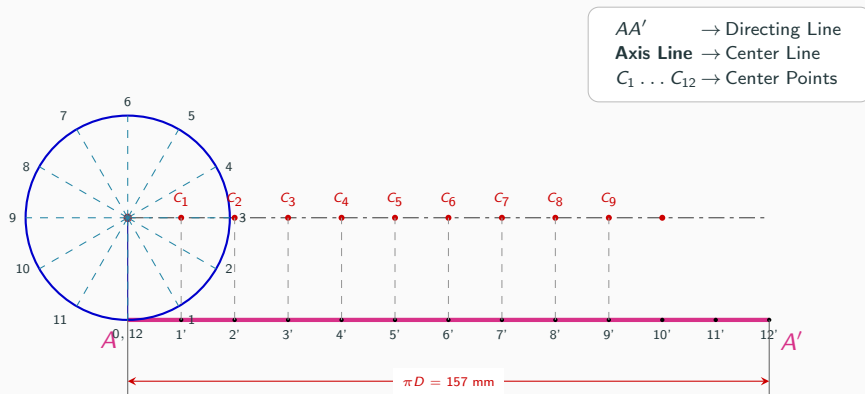


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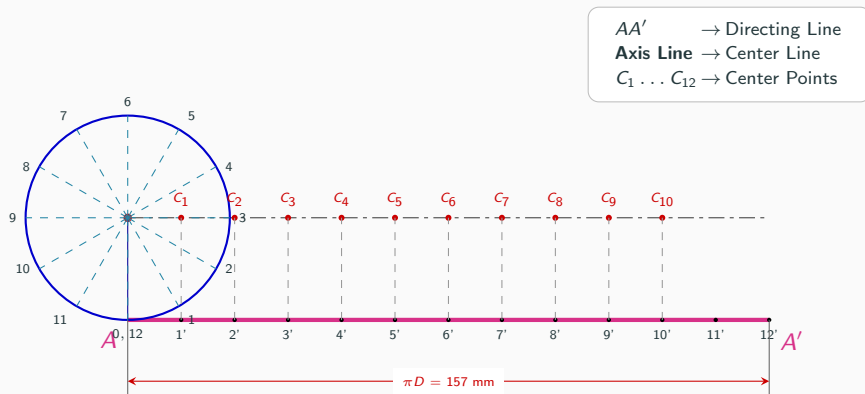




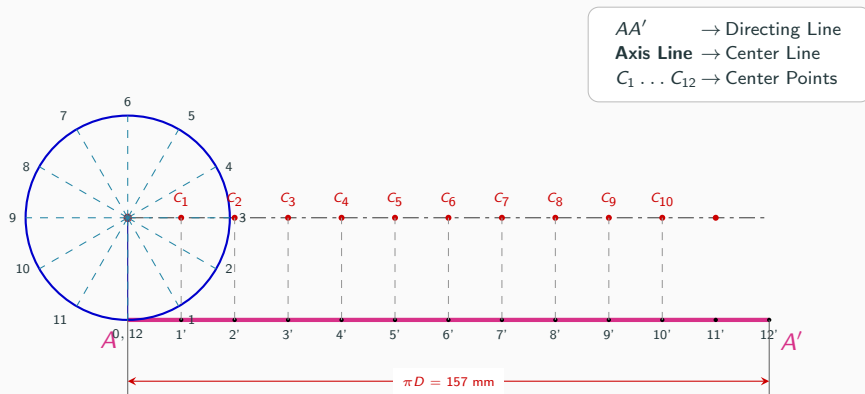
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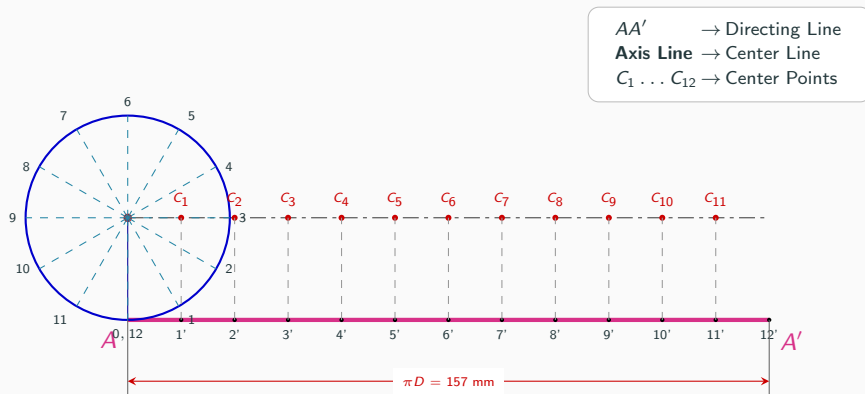
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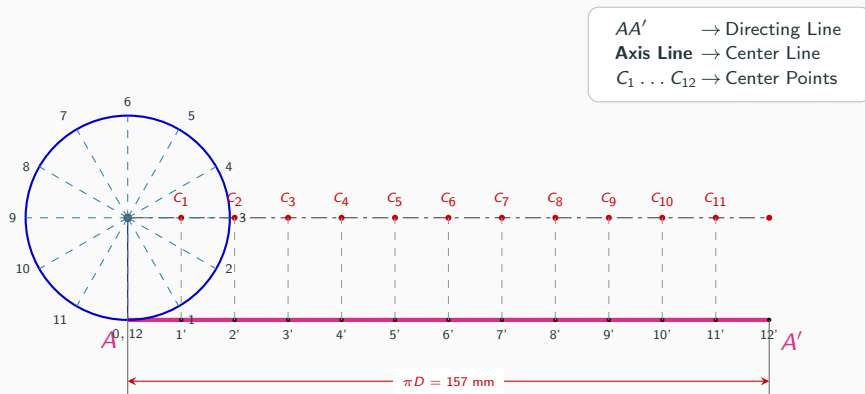
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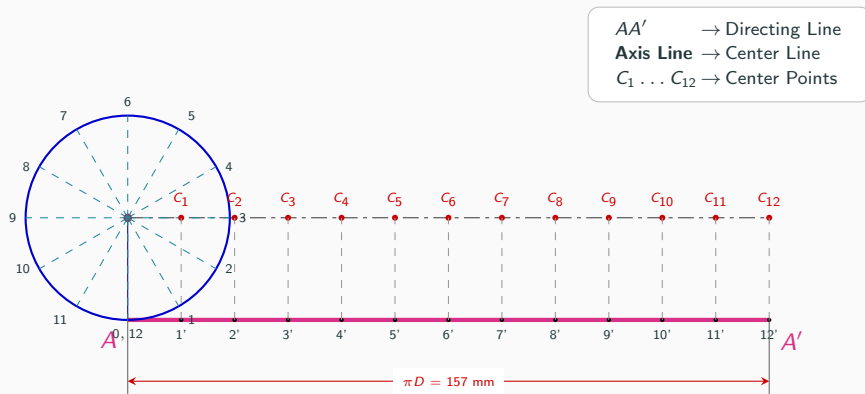
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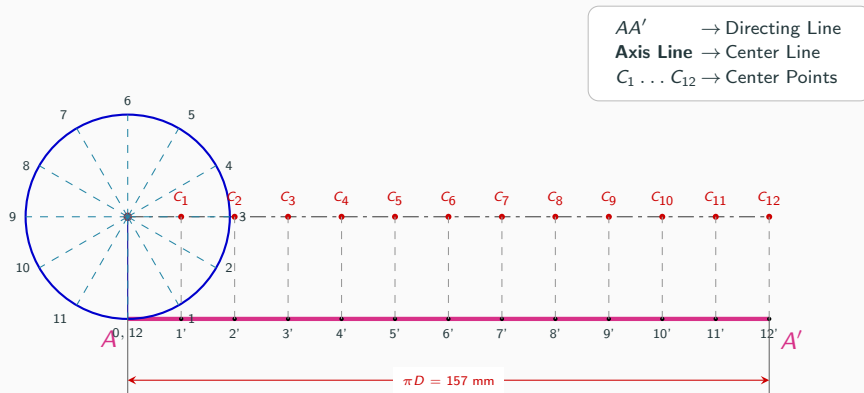
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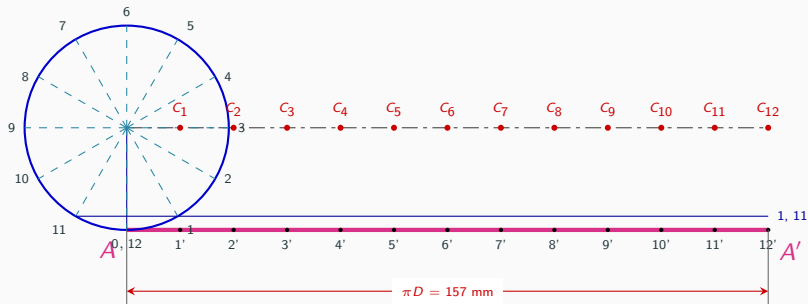
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## Step 5: Draw Horizontal Reference Lines

Blue Lines → Horizontal Reference Lines

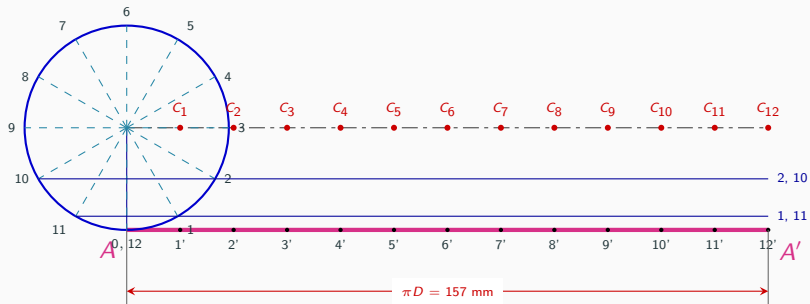
$C_1 \dots C_{12}$  → Retained Centers





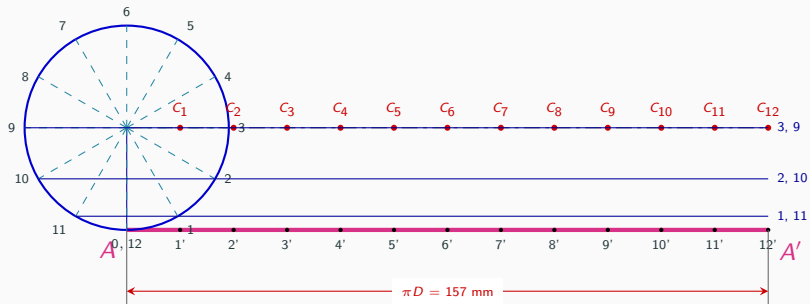
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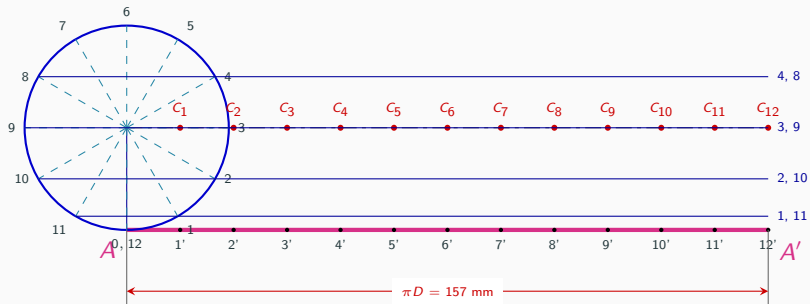
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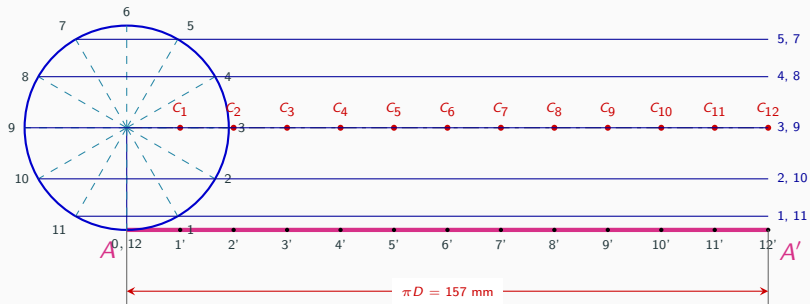
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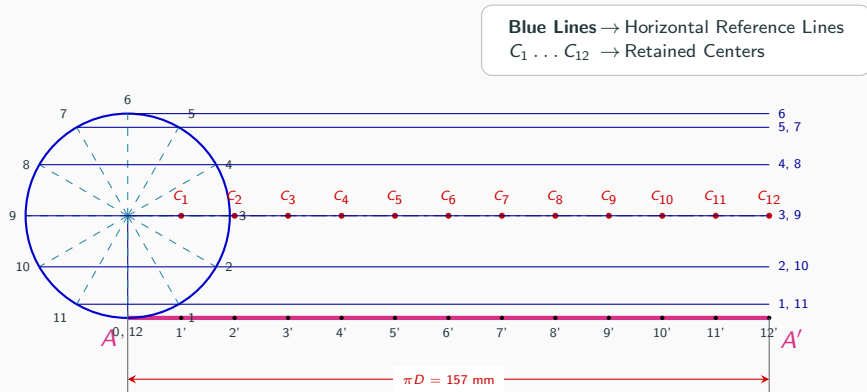
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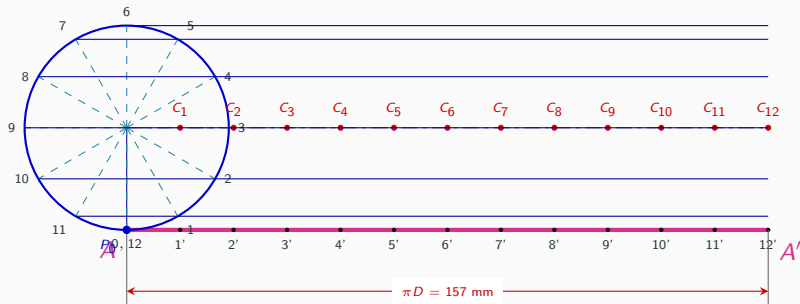


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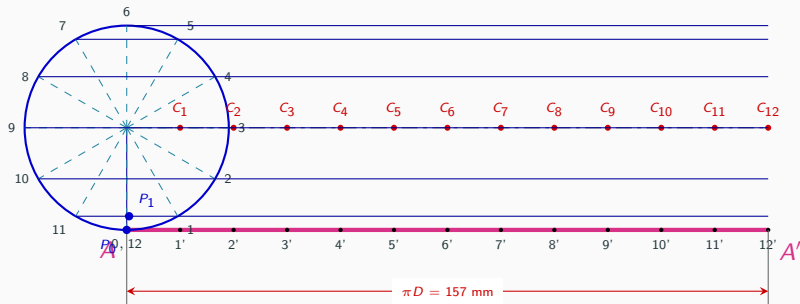
## Step 6: Strike Arcs to Locate Points $P_1 \dots P_{12}$

$C_1 \dots C_{12} \rightarrow$  Retained Centers  
 $P_1 \dots P_{12} \rightarrow$  Cycloid Locus Points



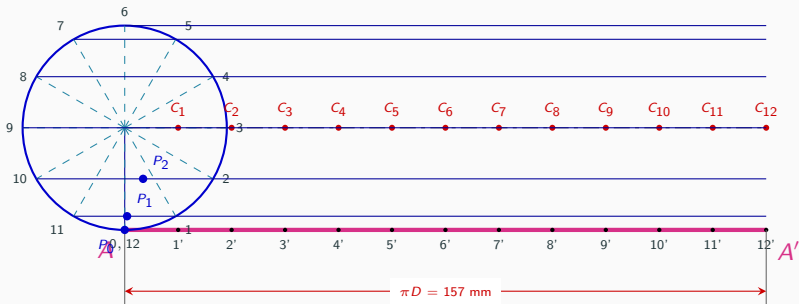
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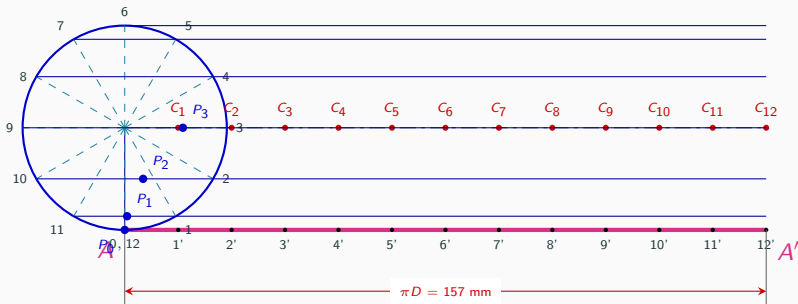
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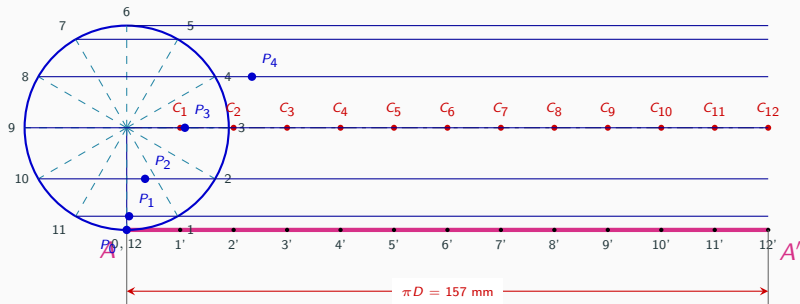
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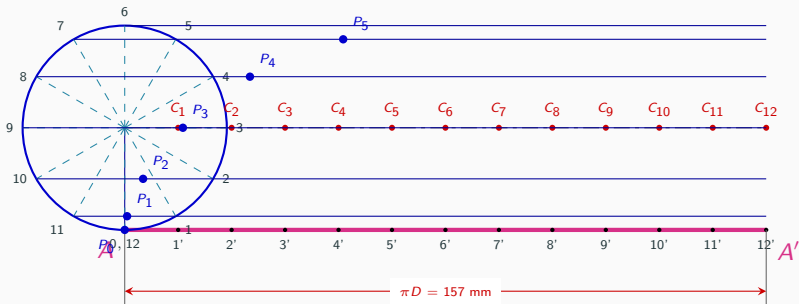
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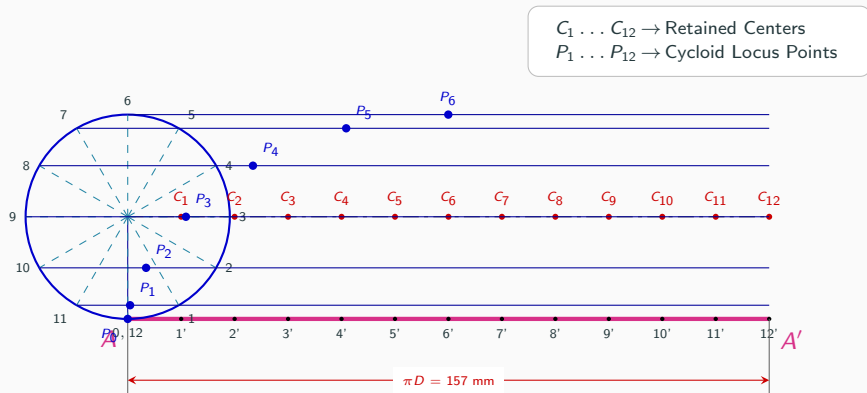


## Step 6: Strike Arcs to Locate Points $P_1 \dots P_{12}$

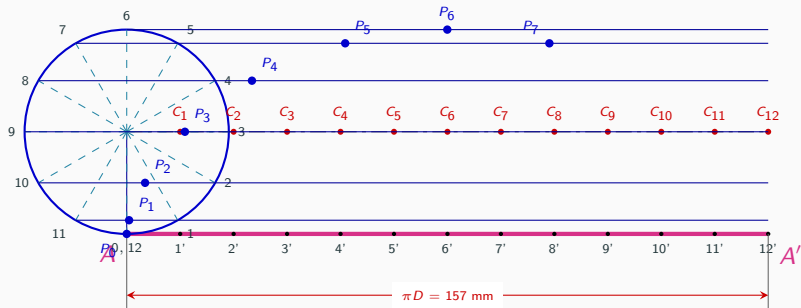
$C_1 \dots C_{12} \rightarrow$  Retained Centers  
 $P_1 \dots P_{12} \rightarrow$  Cycloid Locus Points



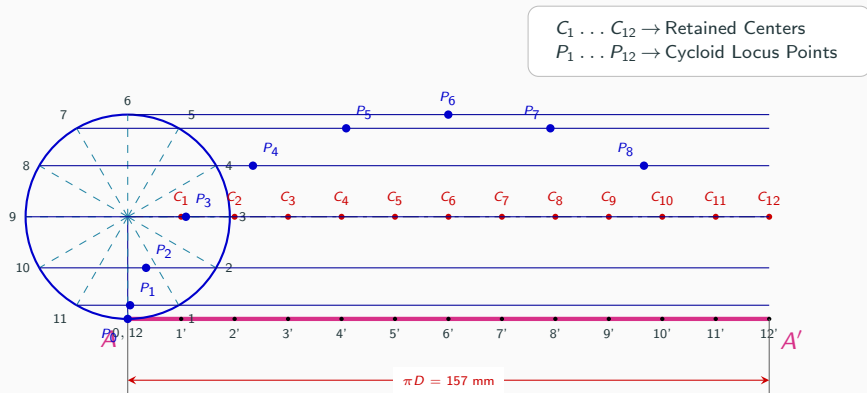
## Step 6: Strike Arcs to Locate Points $P_1 \dots P_{12}$



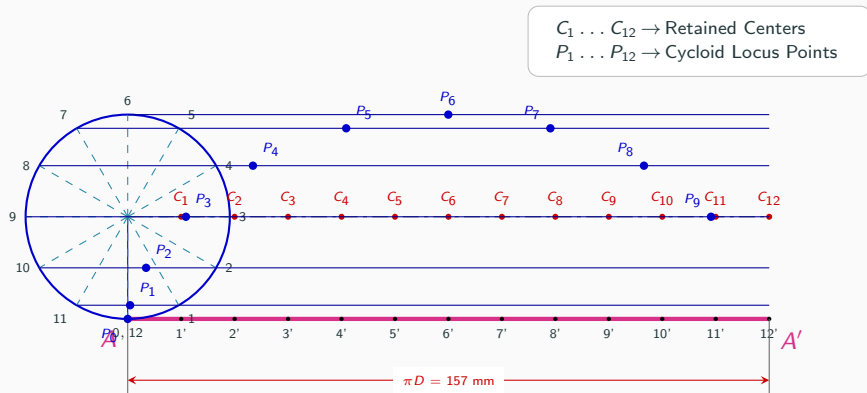
## Step 6: Strike Arcs to Locate Points $P_1 \dots P_{12}$



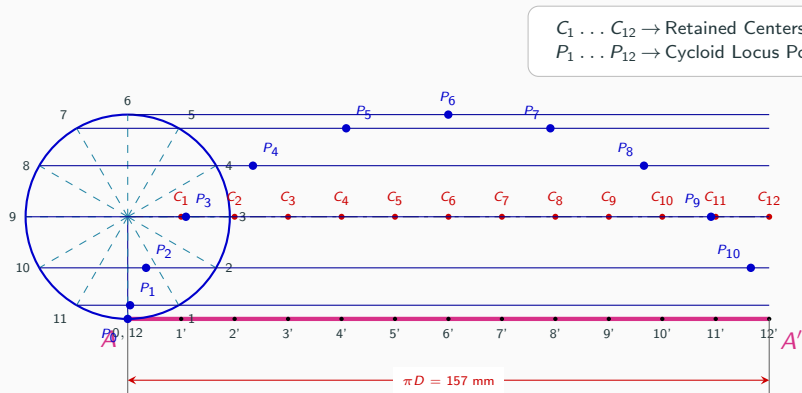
## Step 6: Strike Arcs to Locate Points $P_1 \dots P_{12}$



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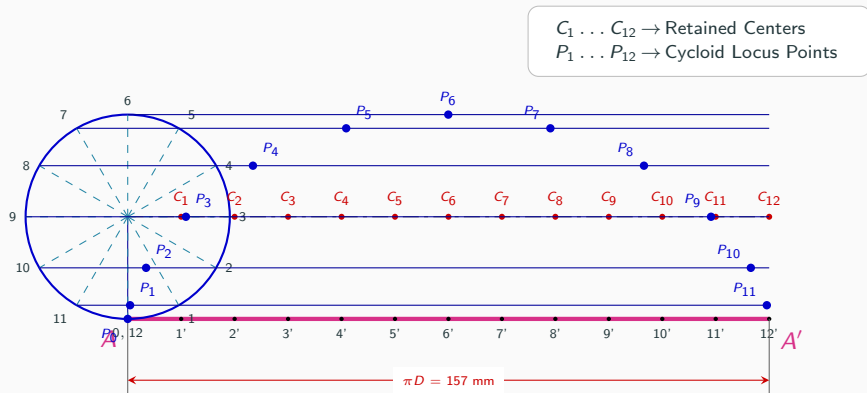


## Step 6: Strike Arcs to Locate Points $P_1 \dots P_{12}$

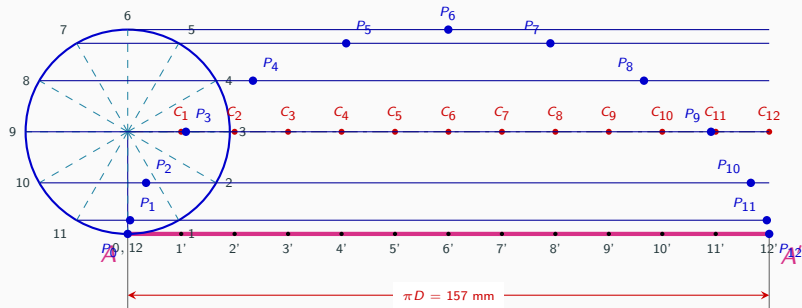




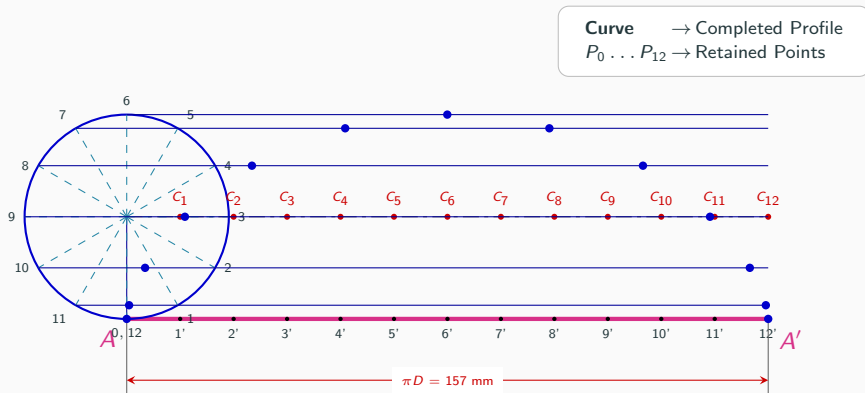
## Step 6: Strike Arcs to Locate Points $P_1 \dots P_{12}$



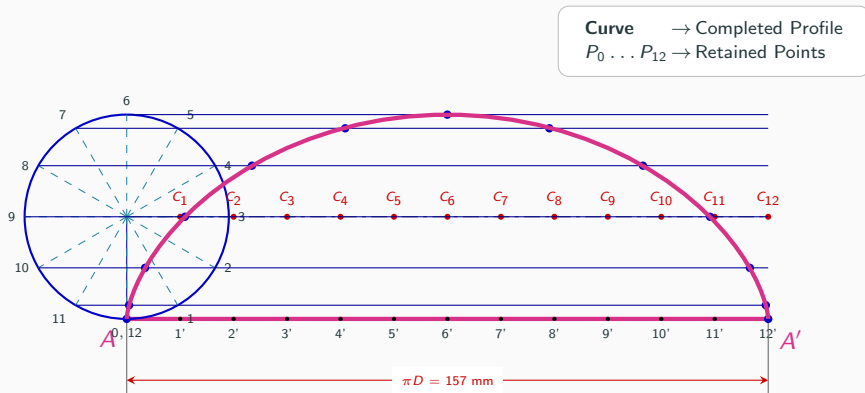
## Step 6: Strike Arcs to Locate Points $P_1 \dots P_{12}$



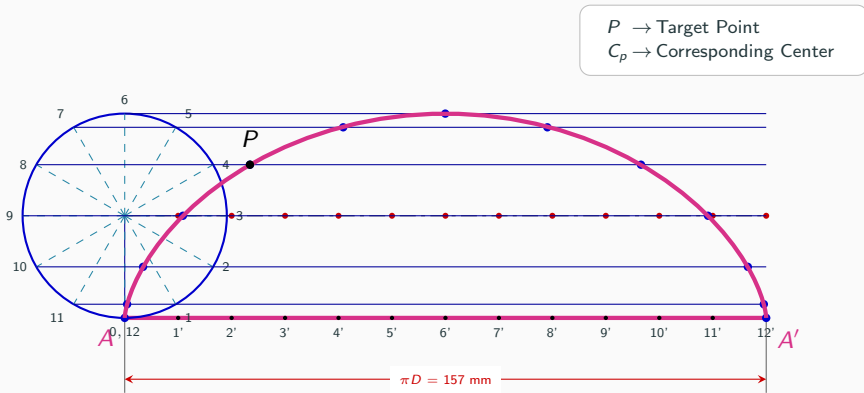
## Step 7: Smoothly Join Points to Form Cycloid Curve



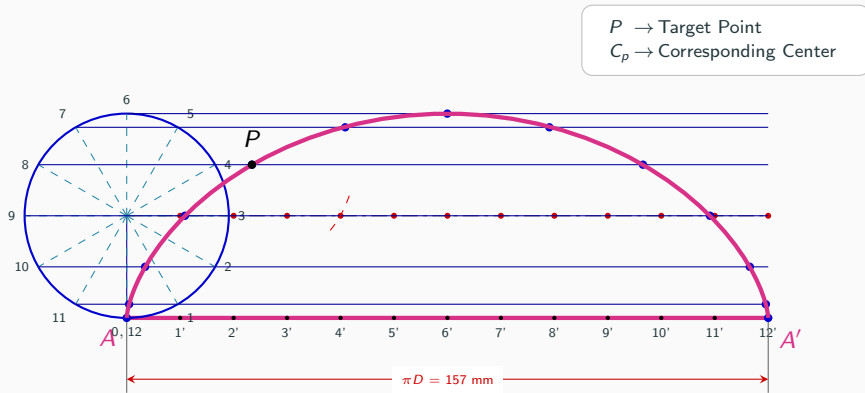
## Step 7: Smoothly Join Points to Form Cycloid Curve



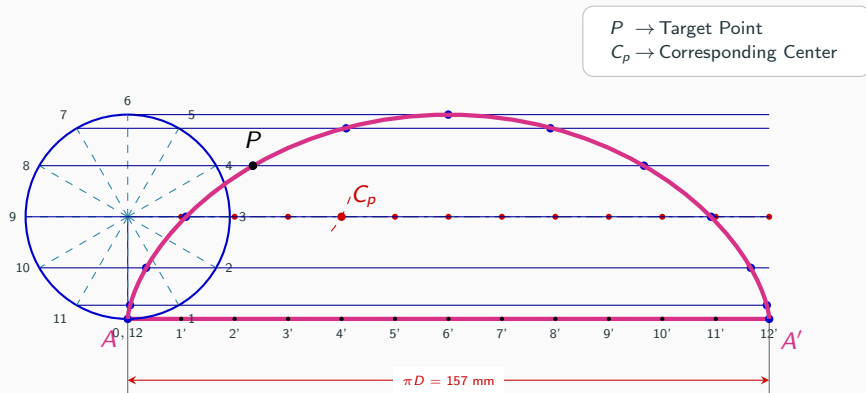
### Step 8: Mark Point $P$ & Locate Center $C_p$



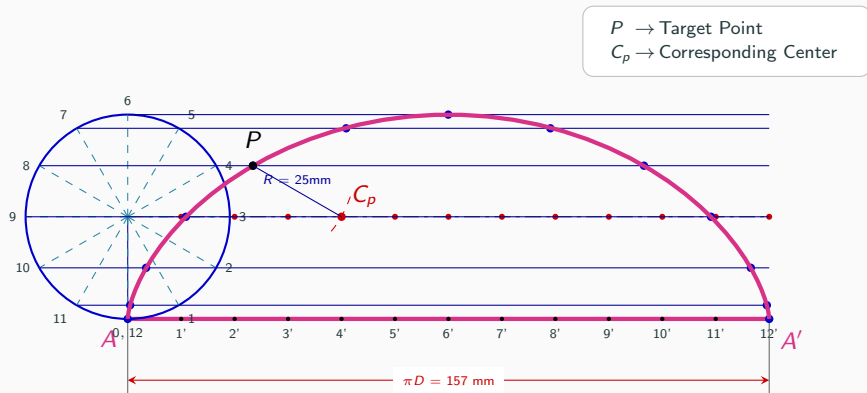
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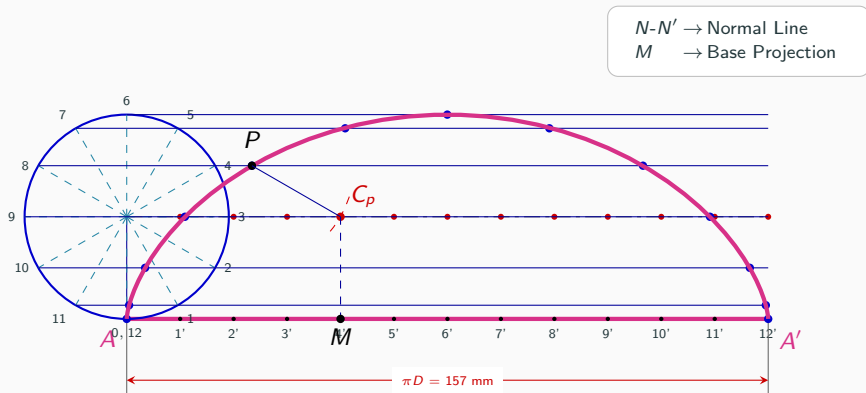


## Step 8: Mark Point $P$ & Locate Center $C_p$

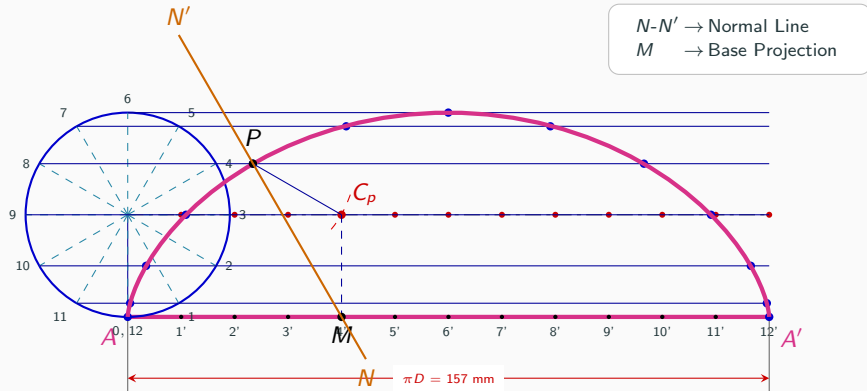




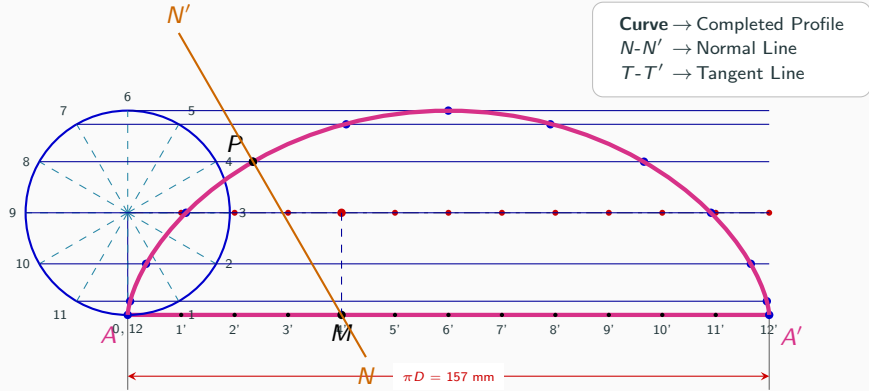
## Step 9: Construct Normal Line $N-N'$



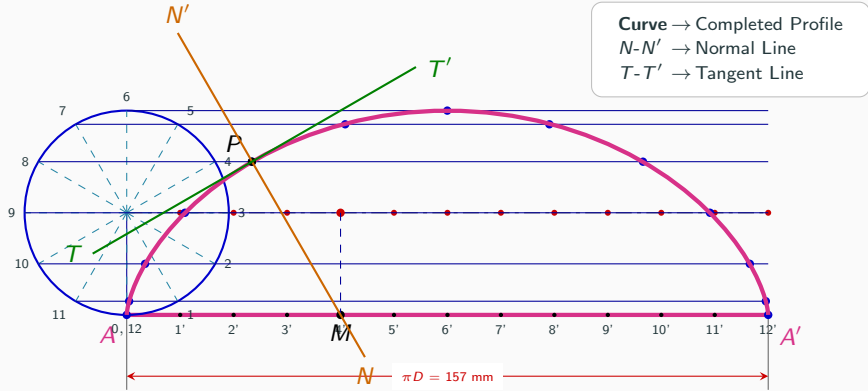
## Step 9: Construct Normal Line $N-N'$



## Step 10: Construct Tangent Line $T-T'$ & Final Result



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